

# Fast Data-Driven Chance Constrained AC-OPF using Hybrid Sparse Gaussian Processes

Mile Mitrovic  
Skolkovo Institute of Science and  
Technology  
Moscow, Russia  
mile.mitrovic@skoltech.ru

Aleksandr Lukashevich  
Skolkovo Institute of Science and  
Technology  
Moscow, Russia  
aleksandr.lukashevich@skoltech.ru

Petr Vorobev  
Skolkovo Institute of Science and  
Technology  
Moscow, Russia  
p.vorobev@skoltech.ru

Vladimir Terzija  
Skolkovo Institute of Science and Technology  
Moscow, Russia  
v.terzija@skoltech.ru

Yury Maximov  
Los Alamos National Laboratory  
Los Alamos NM, USA  
yury@lanl.gov

Deepjyoti Deka  
Los Alamos National Laboratory  
Los Alamos NM, USA  
deepjoyti@lanl.gov

**Abstract**—The alternating current (AC) chance-constrained optimal power flow (CC-OPF) problem addresses the economic efficiency of electricity generation and delivery under generation uncertainty. The latter is intrinsic to modern power grids because of the high amount of renewables. Despite its academic success, the AC CC-OPF problem is highly nonlinear and computationally demanding, which limits its practical impact. For improving the AC-OPF problem complexity/accuracy trade-off, the paper proposes a fast data-driven setup that uses the sparse and hybrid Gaussian processes (GP) framework to model the power flow equations with input uncertainty. We advocate the efficiency of the proposed approach by a numerical study over multiple IEEE test cases that show up to two times speed-up and more accurate solutions over state-of-the-art methods.

**Index Terms**—Power Flow, Chance-Constrained Optimization, Gaussian processes, AC-OPF.

## I. INTRODUCTION

The increase in renewable energy generation is a primary approach to reducing greenhouse emissions. Despite the economic and ecological benefits of renewable energy sources (RES), the uncertainty associated with RES generation, such as wind and solar, leads to significant control challenges for the Transmission System Operators (TSOs) [1]. Optimization and feedback control algorithms should act on second and sub-second levels to prevent the grid from impending energy blackout due to RES uncertainty. Among such fundamental tools, the optimal power flow (OPF) problem is key to finding the economically optimal operation under security constraints. Various methods address the accuracy and scalability of the AC OPF problem under uncertainty, including seminal probabilistic optimal power flow [2], and analytical and scenario-based chance-constrained approaches [3], [4]. The chance-constrained setup considered in this paper ensures that the security constraints are satisfied with a high probability and allows TSOs to balance operating costs and system security. Unfortunately, Chance-Constrained (CC) AC-OPF with nonlinear Alternating Current-Power flow (AC-PF) equations are known to be computationally demanding and often do not

have an analytical solution [3], [5], [6]. A simpler linear version, chance-constraint direct current (DC) approximation (DC CC-OPF) [4], [7], [8] has been widely studied in literature and has a few scalable solutions [9], [10], but suffers from conservative solutions. Furthermore, it requires knowledge of the system parameters, such as line impedances without errors and uncertainties.

The paper proposes to improve the AC CC-OPF scalability using a data-driven method that builds upon the Gaussian Process (GP) Regression framework to replace the power flow equations using prior historical data. Note that GP-based power-flow models have software implementations in deterministic OPF formulations [11], and for CC-OPF in [12], [13]. Compared to vanilla Gaussian process regression in [12], however, our approach uses (1) a **hybrid GP** that combines linear DC-PF with the data-driven estimation of the residuals between DC-PF and AC-PF, and (2) uses **sparse** Gaussian regression using few selective data for the estimation of the hybrid GP. Both these steps significantly reduce our computational complexity without compromising the solution's accuracy, making it amenable for real-time/online deployment. We demonstrate the practical advantage of our approach against state-of-the-art algorithms on multiple IEEE test cases.

This paper is organized as follows. Section II describes the problem addressed and the uncertainty modeling. Section III outlines the Gaussian Process Regression and its sparse version. The hybrid approach to solve the AC CC-OPF problem is given in Section IV. Section V presents numerical evaluation results over various IEEE test cases. Section VI presents a critical discussion, possible extensions, and a summary of the paper.

## II. BACKGROUND AND UNCERTAINTY MODELING

### A. Grid Uncertainty

Let  $\mathcal{Q}(\mathcal{B}, \mathcal{E})$  be a transmission grid with a set of buses  $\mathcal{B}$ ,  $b = |\mathcal{B}|$ , and (directed) lines  $\mathcal{E}$ ,  $e = |\mathcal{E}|$ . We decompose each

power injection  $p_i$  at a bus into controllable generation (referred by subscript  $g$ ), uncertain renewable  $rg$ , and uncertain load  $l$ :

$$p_i = p_{g,i} - p_{l,i} + p_{rg,i}, \quad q_i = q_{g,i} - q_{l,i} + q_{rg,i},$$

where  $p$  and  $q$  are active and reactive power injections resp. The classical AC-OPF problem assumes that deterministic power injections are known; however, in practice, loads and renewable generations fluctuate. Let  $\omega$  be an active power fluctuation so that  $p(\omega) = p + \omega$ ,  $q(\omega) = q + \gamma\omega$  where  $\gamma$  is a constant power factor,  $\gamma > 0$ .

### B. Generation and Voltage Control

Following uncertainty realizations  $\omega$ , controllable generators adapt their generation to maintain the total power balance through the automatic generation control (AGC) [3]. The AGC assumes that the total power mismatch  $\Omega = \sum_{i \in \mathcal{G}} \omega_i$  is divided among the generators according to the participation factors  $\alpha_i$ ,  $\alpha_i > 0$ , as:

$$p_{g,i}(\omega) = p_{g,i} + \alpha_i \Omega, \quad \sum_{i \in \mathcal{G}} \alpha_i = 1, \quad \alpha_i \geq 0, \quad \forall i \in \mathcal{G}. \quad (1)$$

1) *Power Flows*: Assume  $p_{ij}$  and  $q_{ij}$  denote the active and reactive power flows from bus  $i$  to bus  $j$  along line  $(i, j) \in \mathcal{E}$ . In the AC power flow formulation, each transmission line's active and reactive power flows depend non-linearly on the voltage magnitudes  $v$  and voltage angles  $\theta$ . The AC power flow equations are given by:

$$\begin{aligned} p_{ij}(\omega) &= v_i(\omega)v_j(\omega) [G_{ij} \cos[\theta_{ij}(\omega)] + B_{ij} \sin[\theta_{ij}(\omega)]] \\ q_{ij}(\omega) &= v_i(\omega)v_j(\omega) [G_{ij} \sin[\theta_{ij}(\omega)] - B_{ij} \cos[\theta_{ij}(\omega)]] \\ p_i(\omega) &= \sum_{(i,j) \in \mathcal{E}} p_{ij}(\omega), \quad q_i(\omega) = \sum_{(i,j) \in \mathcal{E}} q_{ij}(\omega) \end{aligned} \quad (2)$$

where  $\theta_{ij} = \theta_i - \theta_j$ ,  $G_{ij}$  and  $B_{ij}$  are the entries of the real and imaginary parts of the network admittance.

### C. Chance Constrained AC OPF

The full setup of the AC chance-constrained OPF problem is as follows [3]:

$$\min_{p_g, q_g, v, \theta} \sum_{i \in \mathcal{G}} \mathbb{E}[c_i(p_{g,i}(w))] \quad (3a)$$

$$\text{s.t. } F(\theta(w), v(w), p(w), q(w)) = 0, \quad \forall w \quad (3b)$$

$$\mathbb{P}(p_{g,i}^{min} \leq p_{g,i}(\omega) \leq p_{g,i}^{max}) \geq 1 - \epsilon_p, \quad \forall i \in \mathcal{G} \quad (3c)$$

$$\mathbb{P}(q_{g,i}^{min} \leq q_{g,i}(\omega) \leq q_{g,i}^{max}) \geq 1 - \epsilon_q, \quad \forall i \in \mathcal{G} \quad (3d)$$

$$\mathbb{P}(v_j^{min} \leq v_j(\omega) \leq v_j^{max}) \geq 1 - \epsilon_v, \quad \forall i \in \mathcal{B} \quad (3e)$$

$$\mathbb{P}(s_{ij}(\omega) \leq s_{ij}^{max}) \geq 1 - \epsilon_s, \quad \forall ij \in \mathcal{E} \quad (3f)$$

where  $c_i(\cdot)$  is a cost function of the  $i$ -th generator,  $s_{ij} = \sqrt{p_{ij}^2 + q_{ij}^2}$  is apparent power, and expectations  $\mathbb{E}$  and probabilities  $\mathbb{P}$  are taken over an unknown distribution of uncertainty  $\omega$ . The objective function thus minimizes the expected cost

of active power generation. Equation 3b is composed of AC-PF and AGC from equations 1,2 and, respectively. Technical limits are described in 3c, 3e, 3f and must be satisfied with the prescribed probabilities  $\epsilon_p$ ,  $\epsilon_q$ ,  $\epsilon_v$ , and  $\epsilon_s$ . In this way, the chance constraints define an  $\epsilon$ -reliability set within which an operator can execute additional control actions safely.

This paper considers separable chance constraints, which limit the violation probability for individual constraints and are easy to enforce [3], [5]. The chance constraints (3) are intractable for non-linear AC-PF; however, Gaussian Process-based regression, discussed next, allows designing a more tractable approximation to them. The latter can easily be approached with modern optimization solvers.

## III. GAUSSIAN PROCESS BASED MODELING

### A. Notation

For a matrix  $X$ ,  $X_{ij}$  denotes element  $ij$  of a matrix  $X$ , and  $[x]_i$ ,  $[x]_{:,j}$  its  $i$ -th row and  $j$ -th column, respectively. With  $[x]_{*,j}$ , we denote  $*$  element of  $j$ -th column vector. Further, we will use  $*$  subscript to denote unseen test data. For the given vector  $x$ ,  $\text{diag}(x)$  is used to refer to a diagonal matrix with entries of vector  $x$ , while  $\text{diag}(X)$  represents the vector of diagonal elements of matrix  $X$ . The probability density of  $x$  is denoted as  $p(x)$ ,  $p(x|y)$  representing the probability of  $x$  for given  $y$ .  $\nabla(\cdot)|_{x_*}$  and  $\nabla^2(\cdot)|_{x_*}$  are the first and second derivative of the function  $(\cdot)$  evaluated at  $x_*$ .

### B. Problem Formulation

In the classical AC-OPF modeling with Gaussian processes [11], [12], the Gaussian process regression model infers the full AC-OPF function in Eq. (3) from previously collected input and output measurement samples as

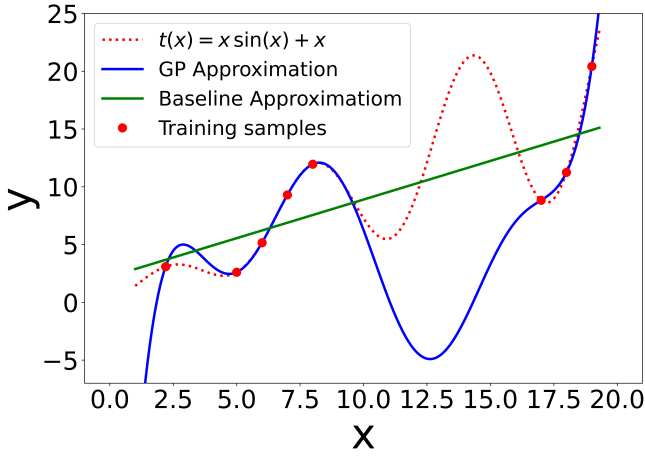
$$y_i = g(x_i) + \delta_i, \quad i = 1, \dots, N$$

where  $N$  is the total number of samples,  $\delta \sim \mathcal{N}(0, \Sigma_\delta)$  is the Gaussian measurement noise with covariance  $\Sigma_\delta$  and  $g$  is the GP function. The input  $x$  is the vector of power injections  $p$ , that includes controllable active power at generators  $p_g$ , and uncertain active power injections  $p_l$  (loads) and  $p_{rg}$  (renewables) (see Eq. II-A). Uncertain reactive power injections depend on active ones via constant power factor as described in Section II-A. The output vector  $y$  corresponds to the rest of the dependent output variables. In short,  $x = [p_g^T, p_l^T, p_{rg}^T]$  and  $y = [v^T, q_g^T, s^T]$ .

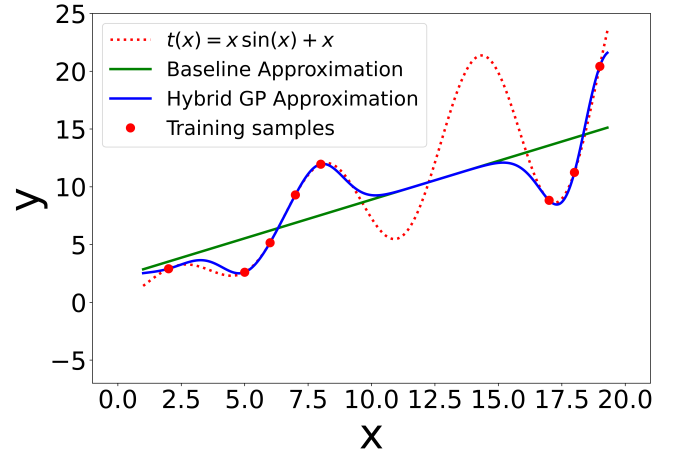
Classical GP regression requires training data to uniformly cover the domain of interest, otherwise, the model can output inadequate prediction. This phenomenon is illustrated in Fig. 1(a). To address this issue, we propose to train GP on residuals between non-linear (AC) and linear (DC) models:

$$r_i = y_i - z(x_i) = g(x_i) + \delta_i, \quad i = 1, \dots, N, \quad (4)$$

where  $z$  is a known or estimated linear part, and  $g$  is an additive GP-term that describes unknown non-linearities of the system. As shown for a candidate non-linear function in Fig. 1(b), this approach uses a linear model as a base and better fits target dependency.



(a) Classical Approach: GP learned on historical data;



(b) Hybrid approach: GP learned on residuals between linear model and data.

Fig. 1. Non-linear function  $x \sin(x) + x$  (in red) is approximated with GP (blue; left) based on several observations (red dots). The hybrid GP approximation (blue; right) uses linear proxy (green) to improve the approximation.

### C. Gaussian Process Regression

Gaussian process regression (GPR) [14] is a non-parametric Bayesian approach used for non-linear regression problems. We employ GPR, as a supervised machine learning model, to infer the vector-valued function  $g(x_i)$  in (4) using a dataset of  $N$  samples, generated from collected measurements of residual outputs  $r_i = g(x_i) + \delta_i$  and inputs  $x_i$ , in set

$$S = \{(x_i, r_i)_{i=1}^N, x_i \in \mathbb{R}^{n_x}, r_i \in \mathbb{R}^{n_y}\}$$

where  $n_x$  and  $n_y$  are the total numbers of input and output variables.

Each output dimension  $r_a$ ,  $a = 1, \dots, n_y$  is learned independently, given the input data  $X = [x_i^T]_{1 \leq i \leq N} \in \mathbb{R}^{N \times n_x}$ . Specifying a GP-prior on  $g$  in each output dimension  $a \in \{1, \dots, n_y\}$  with prior mean function  $m_a(\cdot)$ , chosen kernel  $k_a(\cdot, \cdot)$ , and conditioning on data from set  $S$ , we derive a normally (Gaussian) posterior distribution along dimension  $a$  at a test point  $x_*$ , that is defined with mean and variance [14]:

$$\mu_a(x_*) = k_*^T (K_a + \sigma_{n,a}^2 I)^{-1} [r]_{:,a} \quad (5a)$$

$$\sigma_a^2(x_*) = K_{**} - k_*^T (K_a + \sigma_{n,a}^2 I)^{-1} k_* \quad (5b)$$

where  $\sigma_{n,a}^2$  is a variance of *process* noise corresponding to the *measurement* noise  $\delta$  in (4);  $I$  is an identity matrix;  $[K_a]_{ij} = k_a(x_i, x_j)$  is the Gram matrix of the sampled training data points;  $k_* = k_a(x_i, x_*)$  is the vector of covariance between unseen point and training samples;  $K_{**} = k_a(x_*, x_*)$  is scalar; and  $m_a(X) = [m_a(x_1), m_a(x_2), \dots, m_a(x_N)]$ . We consider a commonly used zero prior mean  $m(X) = 0$  and the squared exponential (SE) kernel for the covariance:

$$k_a(x_i, x_j) = \sigma_{f,a}^2 \exp\left(-\frac{1}{2}(x_i - x_j)^T \Lambda_a^{-1} (x_i - x_j)\right)$$

in which  $\sigma_{f,a}^2$  is the signal variance and  $\Lambda_a = \text{diag}[l_{a,1}^2, l_{a,2}^2, \dots, l_{a,n_x}^2]$  is a positive diagonal length scale

matrix. The predictive distribution of the multivariate output dimension is given by:

$$\begin{aligned} \mu(x_*) &= [\mu_1(x_*), \mu_2(x_*), \dots, \mu_{n_y}(x_*)]^T \\ \Sigma(x_*) &= \text{diag}([\sigma_1^2(x_*), \sigma_2^2(x_*), \dots, \sigma_{n_y}^2(x_*)]) \end{aligned}$$

Cholesky decomposition is applied to minimize computational efforts in inverting  $(K_a + \sigma_{n,a}^2 I)$  during prediction.

GPR training comes down to optimization of the hyperparameters  $\theta = [\{\Lambda_a\}, \sigma_{f,a}^2, \sigma_{n,a}^2]$ , where  $\sigma_{n,a}^2$  is a process noise and  $\{\Lambda_a\}$  is the set of length scale hyperparameters in the symmetric matrix  $\Lambda_a$ . The Maximum Likelihood Estimate with the log marginal likelihood is utilized to optimize hyperparameters as follows [15]:

$$\begin{aligned} \log p([r]_{:,a} | X, \Theta) &= \frac{1}{2} [r]_{:,a}^T (K_a + \sigma_{n,a}^2 I) [r]_{:,a} \\ &\quad - \frac{1}{2} \log |K_a + \sigma_{n,a}^2 I| - \frac{N}{2} \log 2\pi \end{aligned}$$

The negative log-likelihood is minimized to find optimal hyperparameters values using SLSQP solver [16].

### D. Uncertainty Propagation

For given deterministic inputs, GPR model outputs Gaussian distribution with specific mean and variance. However, in this paper, input  $x$  comprising of power injections is stochastic. This makes the output distribution evaluation intractable. To this end, we use Taylor approximation for resulting distribution [17]. To derive that, note that the unseen input  $x_*$  follows

$$x_* \sim \mathcal{N}(\mu_{x_*}, \Sigma_{x_*}) = \mathcal{N}\left(\begin{bmatrix} p_g \\ p_d \end{bmatrix}, \begin{bmatrix} \Sigma_g & \Sigma_{gd} \\ \Sigma_{gd}^T & \Sigma_d \end{bmatrix}\right)$$

where  $p_g, p_d = [p_l^T, p_{rg}^T]^T$  are vectors of optimized generation and forecasted means of uncertain variables (load and renewables), respectively.  $\Sigma_g \in \mathbb{R}^{n_g \times n_g}$ ,  $\Sigma_d \in \mathbb{R}^{n_d \times n_d}$  and  $\Sigma_{gd} \in \mathbb{R}^{n_g \times n_d}$  are the sub-matrices of input covariance matrix.

$\Sigma_d$  models variances of fluctuations of loads and renewable energy and is assumed to be diagonal (uncorrelated fluctuations [4]), so  $(\Sigma_d)_{ii} = \sigma_i^2$ ,  $(\Sigma_d)_{ij} = 0 \forall i \neq j$ . The sub-matrix  $\Sigma_g$  refers to the covariance of fluctuations in generators that arise due to the Automated Generation Control (AGC) (see Eq. 1) and depend on the sum of fluctuations in  $p_d = [p_l^T, p_{rg}^T]^T$ . Following Eq. 1,  $(\Sigma_g)_{ij} = \alpha_i \alpha_j \text{tr}(\Sigma_d)$ ,  $i, j \in \mathcal{G}$ . Similarly, the correlation between generation and the uncertain input  $d$  is given by the sub-matrix  $\Sigma_{gd}$ :

$$(\Sigma_{gd})_{ij} = \alpha_i \sigma_j^2, \quad i \in \mathcal{G}, j \in \mathcal{B} \setminus \mathcal{G},$$

**Output prediction:** To determine the output variable  $a$  ( $\mu_a(x_*)$  and  $\sigma_a^2(x_*)$ ) for the input  $x_* \sim \mathcal{N}(\mu_{x_*}, \Sigma_{x_*})$ , we need to propagate the input uncertainty through Eq. 5. Various approximation methods of the posterior of the GPR model have been proposed in the literature [18], [19]. We consider the first-order Taylor approximation (TA1) for fitting nonlinearities and get the mean value and variance of output distribution as:

$$\begin{aligned} \mu_{TA1}(x_*) &= \mu_a(\mu_{x_*}) \\ \sigma_{TA1}^2(x_*) &= \sigma_a^2(\mu_{x_*}) + \nabla(\mu_{x_*}) \Sigma_{x_*} (\nabla(\mu_{x_*}))^T \end{aligned}$$

Note that under TA1, the predicted mean output for random input  $x_*$  depends only on  $\mu_{x_*}$ . At the same time, the variance additionally takes into account the gradient of the posterior mean (5a) and the covariance  $\Sigma_{x_*}$  of the input of the GPR.

#### E. Sparse Approximation

Large data sets make the GPR model intractable since the inversion of the  $N \times N$  covariance matrix  $K_a$  requires  $O(N^3)$  time. Sparse Gaussian Processes reduce complexity by selecting the inducing inputs and the kernel hyperparameters by maximizing a lower bound of the marginal log-likelihood [20]. The idea behind sparse Gaussian Processes is to use a small set of points, often referred to as inducing points,  $X_m = [x_i^T]_{1 \leq i \leq M} \in \mathbb{R}^{M \times n_x}$ ,  $M \leq N$ , to better approximate marginal likelihood. For such a subsample, the approximate posterior GPR mean and covariance function for output variable  $a$  depends on a mean vector  $\mu_m$  and a covariance matrix  $A_m$  as [20]:

$$\begin{aligned} \hat{\mu}_a(x_*) &= k_{m*}^T (K_{mm,a} + \sigma_{m,a}^2 I)^{-1} \mu_{m,a} \\ \hat{\sigma}_a^2(x_*) &= K_{**} - k_{m*}^T (K_{mm,a} + \sigma_{m,a}^2 I)^{-1} k_{m*} \\ &\quad + k_{m*}^T K_{mm,a} A_m K_{mm,a} k_{m*} \end{aligned}$$

where

$$\begin{aligned} \mu_{m,a} &= \sigma_{m,a}^{-2} K_{mm,a} \Gamma K_{mn,a} [r]_{:,a}, \quad A_m = K_{mm,a} \Gamma K_{mm,a} \\ \Gamma &= (K_{mm,a} + \sigma_{m,a}^{-2} K_{mn,a} K_{nm,a})^{-1} \end{aligned}$$

where  $K_{mm}$  is  $M \times M$  covariance matrix of  $\{X_i\}_{i=1}^M$ ;  $K_{mn}$  is  $M \times N$  covariance matrix between  $\{X_i\}_{i=1}^M$  and the training set  $X$ ;  $k_{m*}$  is the cross-covariance vector.

The sparse approximation reduces the time complexity of the GPR prediction to  $O(NM^2)$  from  $N^3$ , while preserving information from the original sample span.

## IV. HYBRID CC-OPF VIA GAUSSIAN PROCESSES

### A. Setup

In contrast to the classical scenario optimization approach, modeling power flow equations with uncertainty through Gaussian processes leads to a more tractable solution. In our hybrid approach, we substitute the power flow balance equation with a combination of a linear approximation and an additive GPR model trained on the residuals, as discussed in the previous section:

$$\mu_y = f(\mu_{x_*}) + \mu(x_*) \quad (6a)$$

$$\sigma_y^2 = \text{diag}(\Sigma_f(\Sigma_{x_*})) + \text{diag}(\Sigma(x_*)) \quad (6b)$$

where  $f(\mu_{x_*}) = [v_{dc}, A\mu_{x_*}^T + b]$ , with entries for voltage magnitude and other variables (reactive injection and apparent power on lines).  $v_{dc} = 1$  p.u. is the constant DC output voltage,  $A$  and  $b$  are the parameters;  $\text{diag}(\Sigma_f(\Sigma_{x_*}))$  and  $\text{diag}(\Sigma(x_*))$  are column vectors of diagonal matrix elements where  $\Sigma_f = (0, 0; 0A\Sigma_{x_*}A^T)$ .

We now reformulate chance constraints for both output and input decision variables in terms of the uncertainty in the GP functions from the previous section. According to [21], one can express individual output probabilities in terms of the mean  $\mu_y$  and variance  $\sigma_y^2$  as:

$$y_h^{\min} + \tau_{y_h} \sqrt{\sigma_{y_h}^2} \leq \mu_{y_h} \leq y_h^{\max} - \tau_{y_h} \sqrt{\sigma_{y_h}^2} \quad (7)$$

where  $\tau_{y_h}$  is the quantile function  $\Phi^{-1}(1 - \epsilon_{y_h})$  of the standard normal distribution and  $h = 1, \dots, n_y$ . Similarly, one can reformulate constraints on the input variables as in [4]:

$$p_{gk}^{\min} + \tau_{p_{gk}} \alpha_k \text{tr}(\Sigma_d) \leq p_{gk} \leq p_{gk}^{\max} - \tau_{p_{gk}} \alpha_k \text{tr}(\Sigma_d) \quad (8)$$

where  $\tau_{p_{gk}} = \Phi^{-1}(1 - \epsilon_{p_{gk}})$  and  $k = 1, \dots, n_g$ .

The cost function depends quadratically on the participation factors  $\alpha$  and generator active powers  $p_g$ :

$$\begin{aligned} &\sum_{k \in \mathcal{G}} \mathbb{E}_\omega [c_k(p_{g,k}(w))] \\ &= \sum_{k \in \mathcal{G}} \{c_{2k}(p_{gk}^2 + \text{tr}(\Sigma_d)\alpha_k^2) + c_{1k}p_{gk} + c_{0k}\} \end{aligned}$$

where  $\{c_{2k}, c_{1k}, c_{0k}\}_{k=1}^{n_u} \geq 0$  are scalar cost coefficients.

Applying the GP-balance equation (6), an analytical reformulation of chance-constraints with uncertainty margins

(7) (8), the hybrid GP CC-OPF problem (derived from the original Eqs. 3) is given by:

$$\min_{p_g, \alpha, \mu_y, \sigma_y} \sum_{k \in \mathcal{G}} \{c_{2k}(p_{gk}^2 + tr(\Sigma_d)\alpha_k^2) + c_{1k}p_{gk} + c_{0k}\} \quad (9a)$$

$$\text{s.t.} \quad \sum_{k \in \mathcal{G}} \alpha_k = 1, \quad \alpha_k \geq 0 \quad (9b)$$

$$\sum_{k \in \mathcal{G}} p_{gk} = \sum_{i \in n_l} p_{li} - \sum_{j \in n_{rg}} p_{rgj} \quad (9c)$$

$$\mu_y = f(\mu_{x_*}) + \mu(x_*) \quad (9d)$$

$$\sigma_y^2 = \text{diag}(\Sigma_f(\Sigma_{x_*})) + \text{diag}(\Sigma(x_*)) \quad (9e)$$

$$y_h^{\min} + \lambda_{y_h} \leq \mu_{y_h} \leq y_h^{\max} - \lambda_{y_h} \quad (9f)$$

$$u_k^{\min} + \lambda_{p_{gk}} \leq p_{gk} \leq u_k^{\max} - \lambda_{p_{gk}} \quad (9g)$$

where  $\lambda_{y_h} = \tau_{y_h} \sqrt{\sigma_{y_h}^2}$  for  $h \in n_y$  and  $\lambda_{p_{gk}} = \tau_{p_{gk}} \alpha_k \sqrt{tr(\Sigma_d)}$  are the uncertainty margins;  $\sqrt{tr(\Sigma_d)}$  represents the standard deviation of the total active power imbalance  $\Omega$ . Equations (9c)–(9e) stand for approximated AC-PF equations by the hybrid GP.

The proposed OPF optimization method (9) is a non-convex optimization problem. Since the SE kernel is twice differentiable, we adopt the CasADi [22] nonlinear optimization framework using the available IPOPT [23] solver to solve this optimization problem.

### B. Evaluation Metrics

To evaluate the performance of the GPR regression model, the root mean square error (RMSE) metric is used for each output variable  $a$ . If the predicted output vector  $\hat{y}$  (GP-mean values) and ground truth  $y$  (AC-PF output values), the RMSE is calculated as  $RMSE_a = \sqrt{\frac{1}{N_*} \sum_{i=1}^{N_*} ([y]_{i,a} - [\hat{y}]_{i,a})^2}$ , where  $N_*$  is a number of test samples. The average RMSE score of all outputs  $RMSE = n_y^{-1} \sum_{a=1}^{n_y} RMSE_a$  is the accuracy measure we use in our experiments.

## V. RESULTS

We used an Intel Core i7-5500U CPU @ 2.40GHz with 8GB RAM for numerical computations and pandapower [24] package to validate the results. The source code for experiments is accessible online<sup>1</sup>.

### A. Case Study

Performance and scalability of the proposed hybrid sparse GP CC-OPF method are evaluated over 9-bus and 39-bus IEEE test cases. We assume the fluctuations in the system to be 15% of (forecasted) loads ( $\sigma_l = 0.15 p_l$ ) and 30% for the renewable generations ( $\sigma_{rg} = 0.3 p_{rg}$ ). Following [25], we set the acceptable violation probabilities of generator limits to  $\epsilon_{p_g} = 0.1\%$  for controllable generators, and  $\epsilon_y = 2.5\%$  for all other constraints.

<sup>1</sup>[https://github.com/mile888/hybrid\\_gp](https://github.com/mile888/hybrid_gp)

### B. Model Performance

The hybrid and full (no linear component) GPR models are trained on residuals between datasets collected from full AC-OPF and DC-OPF approximation. The synthetic data of the IEEE 9-bus system consists of 8 input features and 15 outputs, while for IEEE 39, the model has 37 inputs and 74 outputs. We generated 75 and 200 random training samples for IEEE 9 and 39-bus systems and considered a different number of inducing points of 10, 30, and 50 for both full and hybrid GPR models. We use IPOPT [23] to solve both hybrid and full GP CC-OPF and find a feasible solution with convergence tolerance of  $\epsilon_{tol} = 10^{-5}$ .

The results in Table I indicate that the hybrid GPR model with TA1 outperforms both RMSE and CPU time with respect to the full GPR model with TA1 in [12]. Similarly, sparse approximation allows to significantly reduce the time complexity for both hybrid and full approaches. In our experiments, we notice that RMSE computed between the DC approximation and the set of samples is  $RMSE = 6.65e^{-2}$  p.u. (IEEE 9) and  $3.04e^{-2}$  p.u. (IEEE 39). In other words, applying the GPR model with a few samples leads to at least 6 times more accurate approximation than the DC baseline.

TABLE I  
GP CC-OPF APPROXIMATION METHOD RESULTS FOR IEEE 9 AND IEEE 39 BUS SYSTEMS.

| Test Case            | IEEE 39                |              | IEEE 9                 |              |
|----------------------|------------------------|--------------|------------------------|--------------|
|                      | RMSE<br>$10^{-2}$ p.u. | Time<br>sec. | RMSE<br>$10^{-2}$ p.u. | Time<br>sec. |
| hybrid TA1           | <b>0.53</b>            | 10.82        | <b>0.30</b>            | 0.24         |
| hybrid-sparse 50 TA1 | 1.12                   | 2.93         | 0.56                   | 0.10         |
| hybrid-sparse 30 TA1 | 1.22                   | 1.23         | 0.79                   | 0.06         |
| hybrid-sparse 10 TA1 | 1.41                   | <b>0.51</b>  | 1.22                   | <b>0.02</b>  |
| full TA1             | 1.37                   | 19.38        | 0.65                   | 0.35         |
| full-sparse 50 TA1   | 2.69                   | 4.12         | 0.97                   | 0.17         |
| full-sparse 30 TA1   | 2.80                   | 1.42         | 1.67                   | 0.13         |
| full-sparse 10 TA1   | 3.25                   | <b>0.51</b>  | 2.43                   | 0.03         |

### C. Comparison of GP CC-OPF to Sample-Based Methods

In this section, we compare optimal objective function value, computational time, and reliability of the proposed hybrid and full GP CC-OPF approaches with Scenario Approximation (SA) CC-OPF [6]. In all cases, we use the first-order Taylor approximation (TA1) as a function of linearization. The empirical constraint violation is computed using 1000 Monte-Carlo samples that follow (Gaussian) uncertainty distribution.

We compare our approach to two baselines. (A) Deterministic AC-OPF problem for each uncertainty realization and taking a corresponding quantile in the output distribution. (B) A deterministic AC-OPF for the mean load and RES, with fixed and equal participation factor  $\alpha^*$ . Notice that (A) is unrealistic as it solves AC-OPF after uncertainty realization (full-resource), while (B) does not optimize the participation factor  $\alpha$ . Thus (A) gives the upper (though unrealistic) bound and (B) gives the lower bound on the feasible solution value. Numerical evaluation of the bounds, given in Table II, shows that the hybrid-sparse GP modeling provides a flexible scheme

that can improve computational effort without sacrificing accuracy.

TABLE II  
COST FUNCTION VALUES, CPU TIME AND PROBABILITY OF A FAILURE FOR VARIOUS SETUPS.

| System Parameters      | IEEE 39      |                  |           | IEEE 9       |                  |           |
|------------------------|--------------|------------------|-----------|--------------|------------------|-----------|
|                        | Cost, $10^6$ | Failure Prob., % | Time sec. | Cost, $10^3$ | Failure Prob., % | Time sec. |
| A (full recourse)      | 7.87         | n/a              | 0.93      | 4.06         | -                | 0.86      |
| B (base case)          | 7.53         | 35.36            | 0.93      | 3.47         | 9.76             | 0.86      |
| Full GP CC-OPF         | 7.75         | 2.36             | 19.38     | 4.04         | 0.44             | 0.35      |
| Hybr. GP CC-OPF        | 7.75         | 2.48             | 9.58      | 4.02         | 0.8              | 0.12      |
| CC-OPF (20 scenarios)  | n/a          | n/a              | n/a       | 3.48         | 7.52             | 4.4       |
| CC-OPF (50 scenarios)  | 7.64         | 14.20            | 96.4      | 3.84         | 4.92             | 12.1      |
| CC-OPF (100 scenarios) | 7.70         | 8.24             | 184.70    | 3.99         | 1.24             | 22.80     |
| CC-OPF (200 scenarios) | 7.81         | 0.16             | 505.00    | n/a          | n/a              | n/a       |

#### D. Comparison of Uncertainty Margins

Figure 2 compares uncertainty margins [26] calculated with the hybrid and full GP CC-OPF (red and blue), the Monte Carlo simulation of AC-OPF for base-case (yellow and green), and the Monte Carlo simulation for CC-OPF solution (brown and salmon) with 100 (IEEE 9) and 200 (IEEE39) scenarios. The bar plots represent an example of voltage, reactive power, and apparent power flow uncertainty margins for each system.

The 3 standard deviations from the mean for uncertainty margins are analytically derived using the estimated GP-variance (9e) around the estimated GP-mean (9d). The uncertainty margins from Monte Carlo are asymmetrical due to MC-based distributions from non-linear AC-OPF. Thus, Monte Carlo uncertainty margins determine upper  $(1 - \epsilon) = 99.73\%$  and lower  $(\epsilon) = 0.27\%$  quantiles (equal to 3 std) of the output distribution around the AC-PF solution  $y(x)$ , denoted by  $y^{1-\epsilon}$  and  $y^\epsilon$ . Thus, the uncertainty margins are:

$$\lambda^{upper} = y^{1-\epsilon} - y(x), \quad \lambda^{lower} = y(x) - y^\epsilon. \quad (10)$$

The power margins are larger in absolute terms than the voltage margins. The GP CC-OPF margins of both approaches are symmetric and comparable to the upper and lower empirical quantiles obtained from the Monte Carlo AC-OPF and CC-OPF simulations in (10). This indicates that the GP uncertainty margins are well approximated.

#### E. Distributional Robustness

In operational practice, a joint distribution of renewables and loads is not known exactly, and one needs to account for it when solving a problem. Furthermore, distribution mismatch might be a consequence of malfunctioned sensors or a cyber-attack. In this section, we show that the proposed approach is robust against a sufficiently large mismatch between the expected distribution and the real one.

Our experiment consists of three major steps: (1) corrupt the data distribution, (2) train over the corrupted data, and (3) test the predictions of models. We use IEEE 9 case with the dataset generated in Section IV for our experiment. In each of the experiments, only one of the buses (bus 4, bus 6, or bus 8) is subjected to crossing out a 15 MW interval, as shown in Fig. 3. Later on, we train the classical and the hybrid GPR

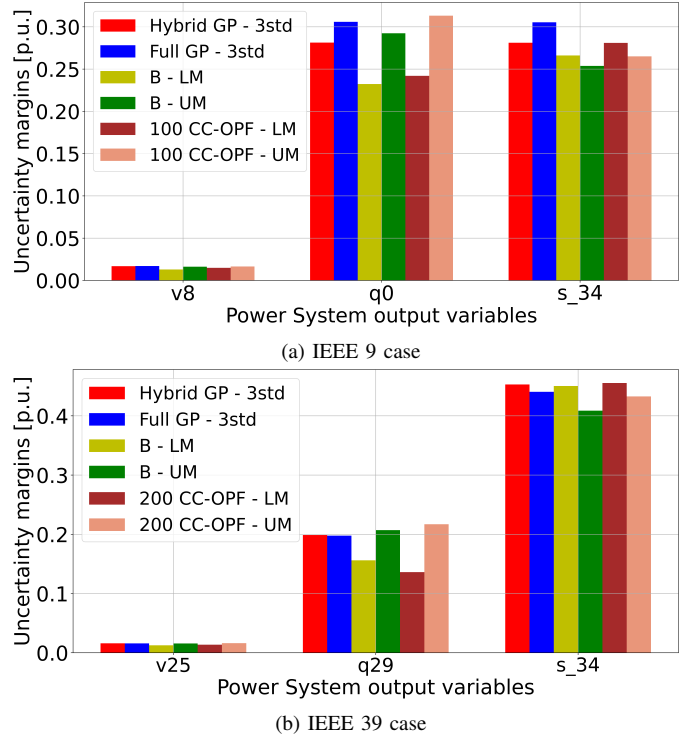


Fig. 2. Output uncertainty margins  $\lambda$ , for example, voltage, reactive and apparent power constraints. (LM - lower margin 0.27% interval; UM - upper margin 99.73% interval)

on this data. Table III illustrates a superior performance of the hybrid GPR in the setup of corrupted training data.

TABLE III  
RMSE COMPARISON FOR GPR AND HYBRID GPR OVER MISSPECIFIED DISTRIBUTIONS. THE LAST COLUMN SPECIFIES THE PORTION OF DATA DISTRIBUTION OF THE CORRESPONDING LOAD THAT HAS BEEN DROPPED.

| Experiment | RMSE GPR             | RMSE Hybrid GPR       | % dropped |
|------------|----------------------|-----------------------|-----------|
| Load 4     | $3.65 \cdot 10^{-1}$ | $9.11 \cdot 10^{-3}$  | 55.0%     |
| Load 6     | $3.89 \cdot 10^{-1}$ | $11.60 \cdot 10^{-3}$ | 38.0%     |
| Load 8     | $0.93 \cdot 10^{-2}$ | $8.34 \cdot 10^{-3}$  | 33.5%     |

## VI. CONCLUSION

The paper studies various machine learning approaches to chance-constrained AC optimal power flow problems. Because of the high non-linearity of the problem, we focused mostly on Gaussian Process Regression, one of the most popular model-free approaches to approximate non-linear stochastic data. As the Gaussian Process Regression has poor scalability, we utilize a sparsity trick that allowed us to significantly reduce the complexity. Finally, we proposed a hybrid GP approach that uses a DC power flow approximation together with the non-linear GP. This demonstrates higher robustness to missing and corrupted data than the classical GP. Overall, this work's findings indicate that the hybrid sparse GP approach that we proposed outperforms other state-of-the-art methods in terms of both speed and accuracy for solving stochastic AC-OPF problems.



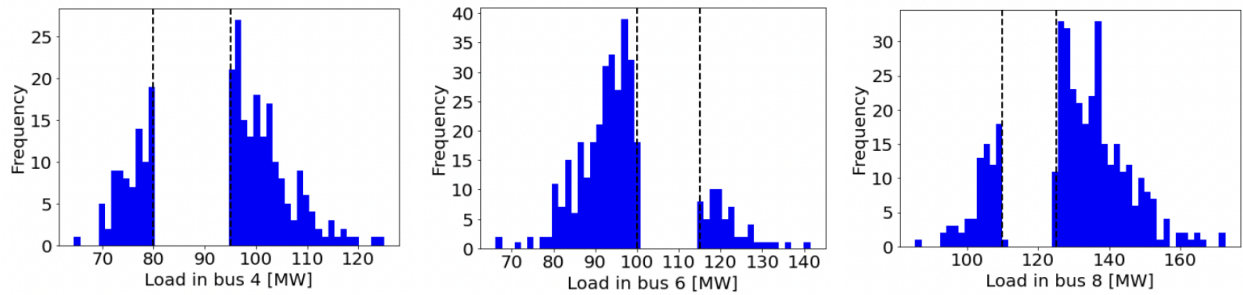


Fig. 3. An example of misspecified input, where a 15 MW interval around the mean bus load value is inaccessible. The latter may happen as a result of cyberattacks or meter failures. The proposed algorithm demonstrates stability over significant adversarial changes in input data.

#### ACKNOWLEDGMENT

Work of M.M., P.V., and V.T. was carried out as a part of the AMPaC Megagrant project, supported by Skoltech (Russia) and the Ministry of Science and Higher Education of Russian Federation, Grant Agreement No. 075-10-2021-067, Grant identification code 000000S707521QJX0002.

Software code implementation was supported by the Russian Science Foundation grant No 22-19-00773, <https://rscf.ru/project/22-19-00773/>.

D.D. and Y.M. acknowledge support from the US Department of Energy's Office of Electricity under the Advanced Grid Modeling (AGM) Program. D.D. acknowledges support from the Directed Research and Development (LDRD) project: "Resilient operation of interdependent engineered networks and natural systems" at LANL.

#### REFERENCES

- [1] D. Koutsoyiannis, "The unavoidable uncertainty of renewable energy and its management," in *Proceedings of the EGU General Assembly Conference, Vienna, Austria*, 2016, pp. 17–22.
- [2] M. J. Morshed, J. B. Hmida, and A. Fekih, "A probabilistic multi-objective approach for power flow optimization in hybrid wind-pv-pev systems," *Applied energy*, vol. 211, pp. 1136–1149, 2018.
- [3] L. Roald and G. Andersson, "Chance-constrained AC optimal power flow: Reformulations and efficient algorithms," *IEEE Transactions on Power Systems*, vol. 33, no. 3, pp. 2906–2918, 2017.
- [4] D. Bienstock, M. Chertkov, and S. Harnett, "Chance-constrained optimal power flow: Risk-aware network control under uncertainty," *Siam Review*, vol. 56, no. 3, pp. 461–495, 2014.
- [5] M. Lubin, Y. Dvorkin, and L. Roald, "Chance constraints for improving the security of ac optimal power flow," *IEEE Transactions on Power Systems*, vol. 34, no. 3, pp. 1908–1917, 2019.
- [6] I. Mezghani, S. Misra, and D. Deka, "Stochastic ac optimal power flow: A data-driven approach," *Electric Power Systems Research*, vol. 189, p. 106567, 2020.
- [7] X. Du, X. Lin, Z. Peng, S. Peng, J. Tang, and W. Li, "Chance-constrained optimal power flow based on a linearized network model," *International Journal of Electrical Power & Energy Systems*, vol. 130, p. 106890, 2021.
- [8] A. Lukashevich, V. Gorchakov, P. Vorobev, D. Deka, and Y. Maximov, "Importance sampling approach to chance-constrained dc optimal power flow," *arXiv preprint arXiv:2111.11729*, 2021.
- [9] N. Viafora, S. Delikaraoglou, P. Pinson, and J. Holbøll, "Chance-constrained optimal power flow with non-parametric probability distributions of dynamic line ratings," *International Journal of Electrical Power & Energy Systems*, vol. 114, p. 105389, 2020.
- [10] A. Lukashevich and Y. Maximov, "Power grid reliability estimation via adaptive importance sampling," *IEEE Control Systems Letters*, vol. 6, pp. 1010–1015, 2021.
- [11] P. Pareek and H. D. Nguyen, "Gaussian process learning-based probabilistic optimal power flow," *IEEE Transactions on Power Systems*, vol. 36, no. 1, pp. 541–544, 2020.
- [12] M. Mitrovic, A. Lukashevich, P. Vorobev, V. Terzija, S. Budenny, Y. Maximov, and D. Deka, "Data-driven stochastic ac-opf using gaussian processes," 2022. [Online]. Available: <https://arxiv.org/abs/2207.10781>
- [13] M. Mitrovic, O. Kundacina, A. Lukashevich, S. Budenny, P. Vorobev, V. Terzija, Y. Maximov, and D. Deka, "Gp cc-opf: Gaussian process based optimization tool for chance-constrained optimal power flow," *Software Impacts*, p. 100489, 2023.
- [14] C. E. Rasmussen, "Gaussian processes in machine learning," in *Summer school on machine learning*. Springer, 2003, pp. 63–71.
- [15] C. K. Williams and C. E. Rasmussen, *Gaussian processes for machine learning*. MIT press Cambridge, MA, 2006, vol. 2, no. 3.
- [16] D. Kraft, "A software package for sequential quadratic programming," *Forschungsbericht- Deutsche Forschungs- und Versuchsanstalt für Luft- und Raumfahrt*, 1988.
- [17] T. Koller, F. Berkenkamp, M. Turchetta, and A. Krause, "Learning-based model predictive control for safe exploration," in *2018 IEEE conference on decision and control (CDC)*. IEEE, 2018, pp. 6059–6066.
- [18] A. Girard, C. E. Rasmussen, J. Quinero-Candela, and R. Murray-Smith, "Gaussian process priors with uncertain inputs? application to multiple-step ahead time series forecasting," in *Advances in neural information processing systems*, vol. 15, 2003.
- [19] M. P. Deisenroth, *Efficient reinforcement learning using Gaussian processes*. KIT Scientific Publishing, 2010, vol. 9.
- [20] M. Titsias, "Variational learning of inducing variables in sparse gaussian processes," in *Artificial intelligence and statistics*. PMLR, 2009, pp. 567–574.
- [21] L. Hewing, J. Kabzan, and M. N. Zeilinger, "Cautious model predictive control using gaussian process regression," *IEEE Transactions on Control Systems Technology*, vol. 28, no. 6, pp. 2736–2743, 2019.
- [22] J. A. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "Casadi: a software framework for nonlinear optimization and optimal control," *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019.
- [23] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical programming*, vol. 106, no. 1, pp. 25–57, 2006.
- [24] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.
- [25] F. Abbaspourtorbati and M. Zima, "The swiss reserve market: Stochastic programming in practice," *IEEE Transactions on Power Systems*, vol. 31, no. 2, pp. 1188–1194, 2015.
- [26] J. Schmidli, L. Roald, S. Chatzivasileiadis, and G. Andersson, "Stochastic ac optimal power flow with approximate chance-constraints," in *2016 IEEE Power and Energy Society General Meeting (PESGM)*. IEEE, 2016, pp. 1–5.